Title:

➢ Real-Time Weather Application using Tkinter

Explanation:

➢ This program is a Graphical User Interface (GUI) based real-time weather intelligence application built using Tkinter, Python's standard GUI library.

➢ The application allows the user to:

- Enter any city name across the world to fetch live weather conditions.
- Click the **"Get Weather"** button or press **Enter** to fetch live meteorological data.
- View real-time temperature, condition description, humidity, and wind speed.
- Receive instant error alerts for invalid city names or network disconnection.

❖ **THE PROGRAM FETCHES REAL-TIME LIVE METEOROLOGICAL DATA FROM OPEN-METEO API AND INCLUDES DYNAMIC WEATHER METRICS.**

Concepts used:

- **tkinter** for GUI design (Tk, Label, Entry, Button, Frame, StringVar)
- **requests** library for making HTTPS REST API calls to Open-Meteo
- **json** parsing to extract temperature, humidity, wind, and weather codes
- **messagebox** for comprehensive error handling and user alerts
- **WMO (World Meteorological Organization)** mapping for standardized conditions

Code:

```
import tkinter as tk
from tkinter import messagebox
import requests

# WMO Weather Interpretation Codes (World Meteorological Organization)
WMO_CODES = {
    0: "Clear Sky ☀️",
    1: "Mainly Clear 🌤️",
    2: "Partly Cloudy ⛅",
    3: "Overcast ☁️",
    45: "Foggy 🌫️",
    48: "Depositing Rime Fog 🌫️",
    51: "Light Drizzle 🌦️",
    53: "Moderate Drizzle 🌦️",
    55: "Dense Drizzle 🌧️",
    61: "Slight Rain 🌧️",
    63: "Moderate Rain 🌧️",
    65: "Heavy Rain 🌧️",
    71: "Slight Snow Fall 🌨️",
    73: "Moderate Snow Fall ❄️",
    75: "Heavy Snow Fall ❄️",
    80: "Rain Showers 🌦️",
    81: "Moderate Showers 🌧️",
    82: "Violent Rain Showers ⛈️",
    95: "Thunderstorm ⚡⛈️",
    96: "Thunderstorm with Hail ⛈️"
}

def get_weather(event=None):
    city = entry_city.get().strip()
    if not city:
        messagebox.showerror("Error", "Please enter a city name.")
        return
```

```
    try:
        # Step 1: Geocoding API to resolve coordinates (Latitude, Longitude)
        geo_url = f"https://geocoding-api.open-meteo.com/v1/search?name={city}&count=1&language=en&format=json"
        geo_resp = requests.get(geo_url, timeout=5)
        geo_resp.raise_for_status()
        geo_data = geo_resp.json()

        results = geo_data.get("results", [])
        if not results:
            messagebox.showerror("Error", f"City '{city}' not found! Please check spelling.")
            return

        loc = results[0]
        city_name = loc.get("name", city)
        country = loc.get("country", "")
        admin1 = loc.get("admin1", "")
        lat = loc["latitude"]
        lon = loc["longitude"]

        # Step 2: Forecast API to retrieve live weather parameters
        weather_url = (
            f"https://api.open-meteo.com/v1/forecast?latitude={lat}&longitude={lon}"
            "¤t=temperature_2m,relative_humidity_2m,apparent_temperature,"
            "weather_code,wind_speed_10m,surface_pressure&timezone=auto"
        )
        weather_resp = requests.get(weather_url, timeout=5)
        weather_resp.raise_for_status()
        weather_data = weather_resp.json()

        curr = weather_data.get("current", {})
        temp = curr.get("temperature_2m", 0.0)
        feels_like = curr.get("apparent_temperature", temp)
        humidity = curr.get("relative_humidity_2m", 0)
        wind = curr.get("wind_speed_10m", 0.0)
        code = curr.get("weather_code", 0)

        condition = WMO_CODES.get(code, "Unknown Condition")
        location_display = f"{city_name}, {admin1}, {country}" if admin1 else f"{city_name}, {country}"

        lbl_location.config(text=location_display, fg="#0F172A")
        lbl_temp.config(text=f"{temp:.1f}°C", fg="#0284C7")
        lbl_condition.config(text=f"Condition: {condition}", fg="#334155")
        lbl_details.config(
            text=f"Feels Like: {feels_like:.1f}°C   |   Humidity: {humidity}%   |   Wind: {wind} km/h",
            fg="#475569"
        )

    except requests.Timeout:
        messagebox.showerror("Error", "Request timed out (5s limit). Please check internet.")
    except requests.ConnectionError:
        messagebox.showerror("Error", "No internet connection. Please verify your network.")
    except Exception as e:
        messagebox.showerror("Error", f"Failed to retrieve weather: {e}")

# --- GUI Setup ---
root = tk.Tk()
root.title("Real-Time Weather Application")
root.geometry("460x420")
root.resizable(False, False)
root.configure(bg="#F1F5F9")

title_lbl = tk.Label(root, text="Real-Time Weather Application", font=("Arial", 16, "bold"),
                     bg="#F1F5F9", fg="#1E293B")
title_lbl.pack(pady=12)

input_frame = tk.Frame(root, bg="#F1F5F9")
input_frame.pack(pady=6)

entry_city = tk.Entry(input_frame, font=("Arial", 12), width=22, relief="solid", bd=1)
entry_city.pack(side="left", padx=6, ipady=3)
entry_city.bind("", get_weather)

btn_search = tk.Button(input_frame, text="Get Weather", command=get_weather,
                       font=("Arial", 10, "bold"), bg="#0284C7", fg="white",
                       relief="flat", cursor="hand2", padx=10, pady=4)
btn_search.pack(side="left")
```

```
card_frame = tk.Frame(root, bg="#FFFFFF", bd=1, relief="solid")
card_frame.pack(pady=16, padx=25, fill="both", expand=True)

lbl_location = tk.Label(card_frame, text="Enter a city to view weather", font=("Arial", 13, "bold"), bg="#FFFFFF",
fg="#64748B")
lbl_location.pack(pady=(16, 4))

lbl_temp = tk.Label(card_frame, text="--°C", font=("Arial", 36, "bold"), bg="#FFFFFF", fg="#CBD5E1")
lbl_temp.pack(pady=4)

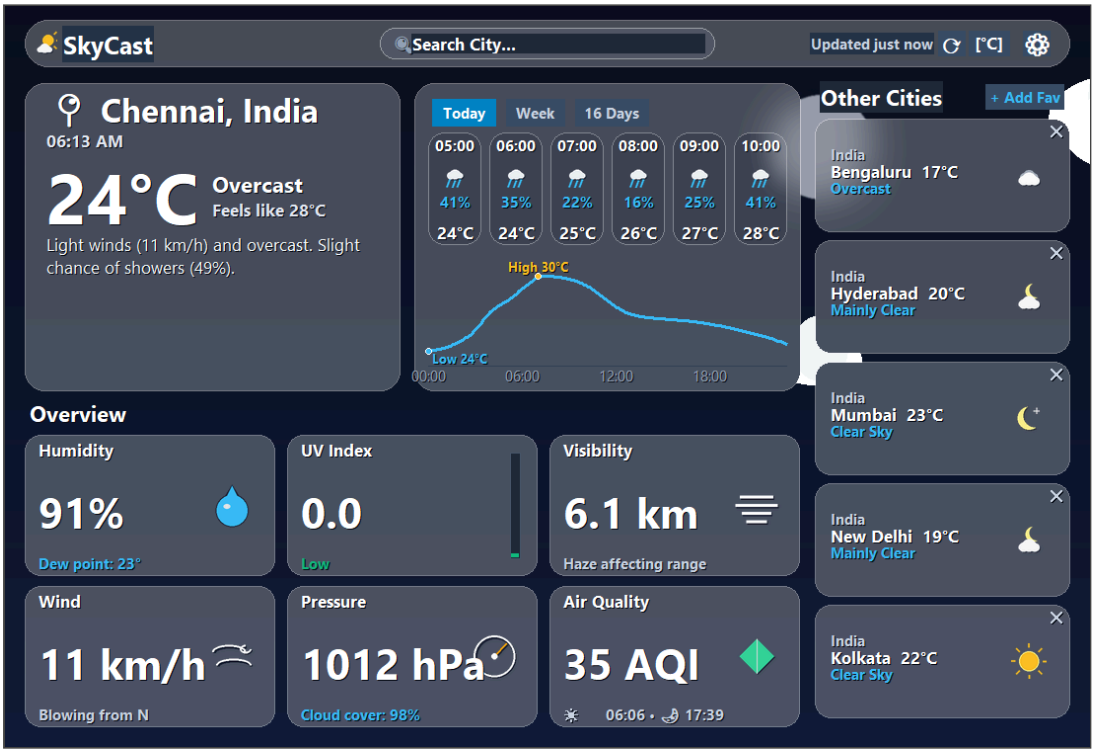
lbl_condition = tk.Label(card_frame, text="", font=("Arial", 12), bg="#FFFFFF", fg="#334155")
lbl_condition.pack(pady=4)

lbl_details = tk.Label(card_frame, text="", font=("Arial", 10), bg="#FFFFFF", fg="#64748B")
lbl_details.pack(pady=(6, 16))

root.mainloop()
```

Output:


EXPLORER
RealTimeWeatherApp
desktop
shared
weather_app.py
requirements.txt
settings.json
weather_app.py
import tkinter as tk
from tkinter import messagebox
import requests
WMO_CODES = {
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3: 'Overcast', 45: 'Foggy', 61: 'Rainy', 95: 'Thunderstorm'
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try:
geo_url = f'https://geocod
geo_resp = requests.get(ge
results = geo_resp.get('re
if not results:
messagebox.showerror('
return
lat, lon = results[0]['lat
Fetch live meteorologica
weather_url = f'https://ap
...
root = tk.Tk()
root.title('Real-Time Weather Appl
root.geometry('500x550')
root.mainloop()
SkyCast
Search City...
Updated just now
Chennai, India
06:13 AM
24°C
Overcast
Feels like 28°C
Light winds (11 km/h) and overcast. Slight chance of showers (49%).
Today
Week
16 Days
Other Cities
Overview
Humidity
91%
Dew point: 23°
UV Index
0.0
Low
Visibility
6.1 km
Haze affecting range
Wind
11 km/h
Blowing from N
Pressure
1012 hPa
Cloud cover: 98%
Air Quality
35 AQI
06:06 • 17:39


SkyCast
Search City...
Updated just now
Chennai, India
06:13 AM
24°C
Overcast
Feels like 28°C
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